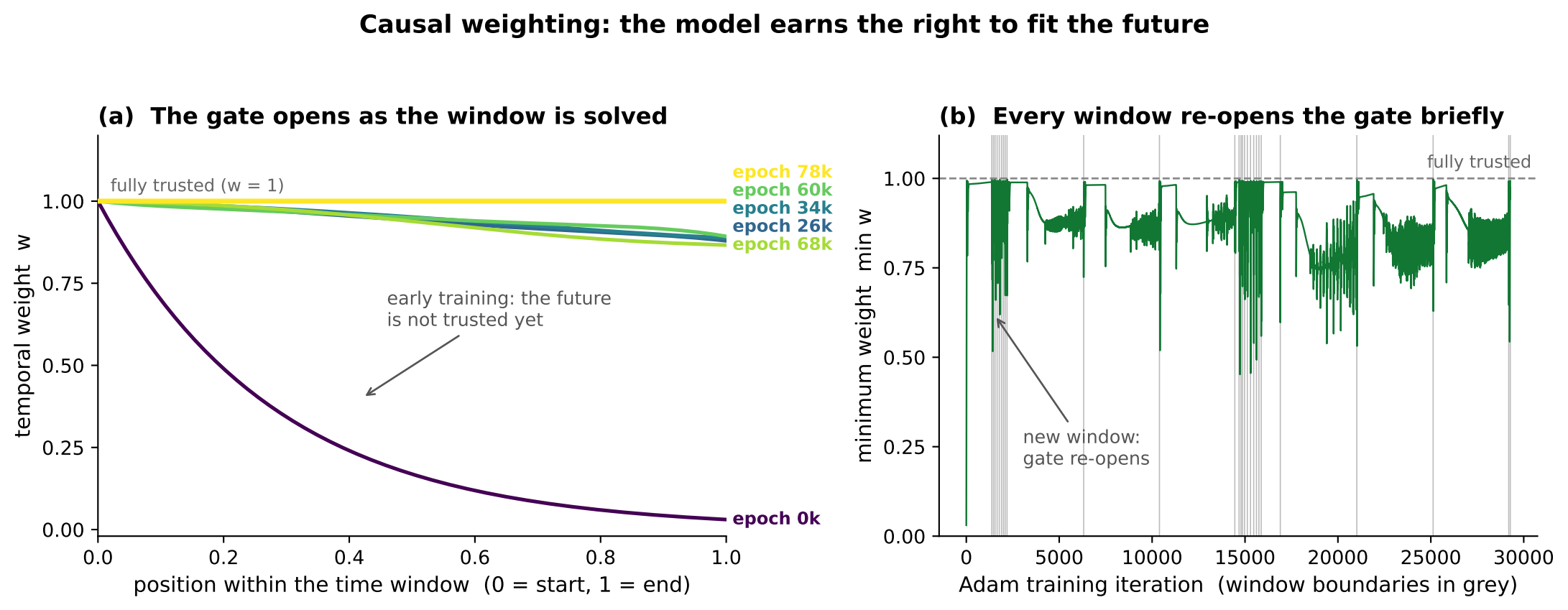

Causal weighting: the model earns the right to fit the future
(a) The gate opens as the window is solved
fully trusted (w = 1)
epoch 78k
epoch 60k
epoch 34k
epoch 26k
epoch 68k
early training: the future
is not trusted yet
epoch 0k
temporal weight w
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
position within the time window (0 = start, 1 = end)
(b) Every window re-opens the gate briefly
fully trusted
new window:
gate re-opens
minimum weight min w
1.00
0.75
0.50
0.25
0.00
0
5000
10000
15000
20000
25000
30000
Adam training iteration (window boundaries in grey)